

Requests to wolfSSL

Lightrider Inc, Veloce PQC SDK. Certificate #4718, module v5.2.1, bundle wolfssl-5.9.2-commercial-fips-linuxv5.2.1, Agreement v.01-24.

1 Our configuration, for your verification

Linux build	<code>./configure --enable-fips=v5</code> ; make, fips-hash.sh, make, testwolfcrypt. No <code>--enable-wolfEntropy</code> , per your 2026-08-27 guidance that wolfEntropy was never tested with v5.2.1.
Windows build	UG 2.2.45 (OE49): wolfssl-fips.sln, DLL Release x64, Visual Studio 2026, no PAA, your supplied <code>user_settings.h</code> (HAVE_FIPS_VERSION 5 / MINOR 2 / PATCH 1).
Seeding	Custom generate-seed callback registered with <code>wc_SetSeed_Cb</code> (FIPS FAQ §2): reads OS kernel entropy (<code>getrandom</code> on Linux, <code>BCryptGenRandom</code> on Windows), applies SP 800-90B-style RCT (cutoff 31) and APT (325/512) to every seed block, fails closed on any error. Documented under legacy IG 9.3.A; no ESV certificate claimed; caveat “no assurance of the minimum strength of generated SSPs” acknowledged.
Distribution	wolfCrypt as object code only inside Veloce installers; notices retained; source confined to a restricted private repository.

2 Please verify or confirm

1. The Linux and Windows build configurations above are approved configurations for module v5.2.1 under the User Guide / COGM.
2. The seeding design above raises no concern for v5.2.1 under legacy IG 9.3.A.
3. Authenticode signing of the FIPS DLL or of the MSI installer does not interfere with the in-core HMAC-SHA-256 integrity check.

3 Please notify us

1. When **OE49** (Windows 11 Pro, Lenovo ThinkPad T14, Intel i7-1260P) and **OE36** (Ubuntu 24.04 on Docker 28.5 / Hyper-V, Xeon Platinum 8272CL, Azure Standard_D2s_v4; CAVP validation 36918) appear on the published #4718 certificate and security policy, with any expected CMVP timeframe. Until publication we ship Windows with validation status “pending publication”.

4 Please advise on entropy source validation

Our V1 seed source is OS kernel entropy and is not ESV-capable: `getrandom` and `BCryptGenRandom` return conditioned output, not raw noise. Under IG 9.3.A Additional Comment 11 an ESV certificate counts only when its number appears on the module certificate and in the security policy, so any ESV credit for Veloce depends on the module side.

1. The ESV certificate and operating-environment list that will back the v5.2.4 module and the FIPS v7 “PQ-FS” module, and whether Windows 11 x86-64 and Ubuntu 24.04 x86-64 will be covered. ESV #E335 currently lists one OE (AEDS Software Distribution 2.0, Xeon E3-1275 v6) with reuse restricted to vendor.
2. Whether wolfSSL will cite a third-party ESV certificate (a Lightrider-owned entropy source or an IDQ Quantis QRNG) in a wolfCrypt module security policy under a commercial engagement, or alternatively offer an OEM module validation for Lightrider.
3. Whether the in-process resubmission of #4718 for OE49 and OE36 changes the module’s legacy IG 9.3.A status or its entropy caveat.

5 Please paper on the Order Form

1. Veloce as the named Program under §3: a PQC SDK exposing cryptographic services through a local agent, not a linkable cryptography library.
2. Commercial coverage for our PQC provider: standalone compilation of `wc_mlkem.*`, `wc_mldsa.*`, and `sha3.c` from the public v5.9.2-stable tree (the commercial bundle strips SHAKE), shipped as object code only.
3. Budget and timing for the FIPS v7 “PQ-FS” upgrade (§10 treats it as an upgrade requiring a new Order Form).